Sudhanva Devanathan

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Experience

BlackRock — Mumbai, India
Software Engineer II — *Jan 2021 – July 2024*

- Built a **Python** orchestration pipeline from scratch on the Aladdin platform processing **100M+ rows** nightly, delivering analytics across **100K+ portfolios** for **130+ clients** to enable a new product offering.
- Reduced concurrent and total daily DB connections by **80%** by building a **Java/Spring Boot** configuration microservice with **Kafka**, **gRPC**, and **Protobuf** to decouple downstream DB dependencies.
- Increased on-time failure detection by **40%** across critical financial models by building an event-driven monitoring system using **asyncio** in **Python** with distributed servers/pollers, surfacing live failure data to an operations dashboard.
- Raised code coverage from **80% to 95%** across **30% of team-owned repos** by migrating **CI/CD** pipelines from **Jenkins** to **Azure DevOps**.
- Migrated a legacy **Perl** process to a config-driven async **Python** service, reducing DB queries by **30%** and cutting average client execution time by **30%**.

BlackRock — Mumbai, India
Software Engineer I — *July 2018 – Dec 2020*

- Reduced analytics job latency by **20%** by building a memory-aware scheduling algorithm that routed jobs to nodes based on historical memory usage, unblocking high-load workloads.
- Maintained **98% availability** across risk-analytics services as **Level 1 on-call engineer** by diagnosing and resolving infrastructure incidents in real-time during a period of high system volatility.
- Shipped **5+ features** into a shared **Go** orchestration engine and its custom DSL, enabling teams to build data pipelines for a parallel rewrite of the legacy **C++** Risk Calculator.

Projects

Distributed Key-Value Store – *Go, Raft, gRPC, Protobuf, AWS* — April 2026

- Built a strongly consistent, fault-tolerant key-value store in **Go** across a 5-node cluster, implementing the **Raft** consensus algorithm from scratch for leader election and log replication with no third-party consensus library.
- Engineered a persistent storage layer from scratch using a **Write-Ahead Log (WAL)** and **LSM Tree** with MemTable, SSTable, Bloom filters and multi-level compaction which achieves crash recovery under **2 seconds** and a **60% reduction** in memory footprint.
- Implemented **consistent hashing** with virtual nodes for data sharding across the cluster, mirroring Cassandra's architecture to minimize data movement on node joins and failures.

Chord DHT with Probabilistic Filter Routing – *C++11* — April 2026

- Designed a **Chord DHT** with three pluggable routing strategies (finger table baseline, **Cuckoo filter**, **Quotient filter**); bridged the **nearest-predecessor vs membership** query mismatch via a 1024-bin spatial index that scans the filter backward from the target key.
- Achieved **O(log n)** lookup hop count for both filter variants while reducing maintenance-message overhead by roughly **50%** during stable periods, since filters skip the continuous `fix_fingers` background repair the baseline requires.
- Built a dual transport layer (**SimTransport** in-process, **IPCTransport** over Unix sockets) sharing one node binary; benchmarked hop count, latency, decision time, and FP sensitivity across ring sizes from 8 to **1024 nodes** running as separate OS processes.

Education

Northeastern University — Boston, MA
Master of Science in Computer Science; GPA: 3.84 — *Sep 2024 – May 2026*

Technical Skills

Languages: Go, Java, Python, C++, SQL, Shell
Systems: Distributed Consensus (Raft), Log-Structured Storage (LSM Tree, WAL, SSTable), Consistent Hashing, Concurrency, Microservices, Event-Driven Architecture
Frameworks & Tools: Spring Boot, Hibernate, gRPC, Protobuf, Kafka, Redis, Docker
Cloud & DevOps: AWS, Azure DevOps, Jenkins, CI/CD, Kubernetes
Databases: PostgreSQL, Redis, Cassandra